

SNEHIT SHARMA

Software Engineer | Backend Developer | Python | FastAPI | PostgreSQL | Redis | AI/LLM
Chennai, India | snehitsharma@outlook.com | +91 8667221735 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of Massachusetts, Boston, USA - M.Sc. in Information Technology | GPA 3.9/4.0 Jan 2023 – Dec 2024
SRM Institute of Science and Technology, IN - B.Tech. Computer Science and Engineering | 74.82% Jul 2017 – May 2021

WORK EXPERIENCE

AI Engineer | Quantile Tech | Boston, US Feb 2025 – Present

- Implemented Protobuf serialization over persistent WebSocket connections to async FastAPI ingestion services, including reconnection and client-side buffering to handle unreliable connections.
- Implemented bounded Redis Streams with consumer workers and backoff signaling, helping the real-time ingestion pipeline absorb bursty traffic, sustaining up to 1,200 events/sec into PostgreSQL/TimescaleDB without data loss.
- Worked on Redis queue + worker-pool decoupling of ingestion from database writes, with timestamp/sequence-based event reordering to handle out-of-order data.
- Built an anomaly-triggered RAG diagnostic pipeline using embeddings, retrieval, LangChain, and historical timeseries context to generate evidence-grounded diagnostic reports, reducing manual investigation time by 25%.
- Implemented automated RAGAS + DeepEval evaluation against synthetic ground truth, improving context precision by ~5% and faithfulness by 12%.

Data Engineer (Assistant System Engineer) | Tata Consultancy Services | Bangalore, IN May 2021 – Jan 2023

- Built and optimized large-scale PySpark/Databricks ETL pipelines for customer transaction data, implementing column masking to support data-protection compliance at scale.
- Owned the FastAPI data-access layer and optimized PostgreSQL queries using execution plans and CTEs, reducing query time by 30% and improving retrieval throughput by 40%.
- Containerized and deployed the data service using Docker/Kubernetes, implementing zero-downtime rolling upgrades and reducing maintenance downtime by 80%.
- Replaced ad-hoc SQL requests and manual data extracts with self-serve REST APIs, backed by database indexing and query optimization, reducing internal report load times and manual data-pull requests.

PROJECTS

GraphRAG & Multi-Source Retrieval Engine (Personal / open-source) | [github](#)

Python · FastAPI · Qdrant · Neo4j · Groq Llama 3.3 70B / Llama 3.1 8B · Gemini embeddings · LangGraph · Docker

- Engineered hybrid graph-vector retrieval system integrating Confluence, Notion, and Slack as a unified knowledge graph, Groq Llama 3.1 for ingest-time entity and relationship extraction to enable relationship-aware search.
- Designed a multi-stage query pipeline spanning semantic caching, query rewriting, entity extraction, 2-hop graph traversal, hybrid search, cross-encoder reranking, and guardrails, feeding Groq Llama 3.3 70B for generation and improving context precision by ~20% over standalone vector search.
- Added Redis/Qdrant-backed semantic caching with query rewriting, reducing average response latency by ~25%, and fully containerized the system with Docker Compose for production-style deployment.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript/TypeScript

Backend & APIs: FastAPI, REST APIs, WebSockets, Protobuf

Databases & Infrastructure: PostgreSQL, TimescaleDB, Redis, Qdrant, Neo4j, Docker, Kubernetes, Redis Streams

AI/LLM: RAG, GraphRAG, LangChain, LangGraph, RAGAS, DeepEval

Data Engineering: PySpark, Databricks, ETL, Time-Series